

Digital Signal Processing Lab BSAI – 7th Semester

Lab 6: Linear Phase FIR Filters

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Objective

The objective of this lab is to study and analyze linear phase FIR filter structures. The main goals include:

- Studying linear phase FIR filter properties.
- Implementing Type-I and Type-III FIR filters.
- Designing an ideal low-pass FIR filter.
- Analyzing magnitude, phase, and group delay responses.
- Identifying FIR filter types using symmetry properties.

MATLAB Function: Hr Type-I

Description: function computes the amplitude response of a Type-I linear phase FIR filter using cosine expansion based on symmetry of the impulse response.

```
1 function [Hr,w,a,L]= Hr_Type1(h)
2
3 M = length(h);
4 L = (M-1)/2;
5 a = [h(L+1) 2*h(L:-1:1)];
6 n = 0:L;
7 w = (0:1:500)*pi/500;
8 Hr = cos(w'*n)*a';
9 end
```

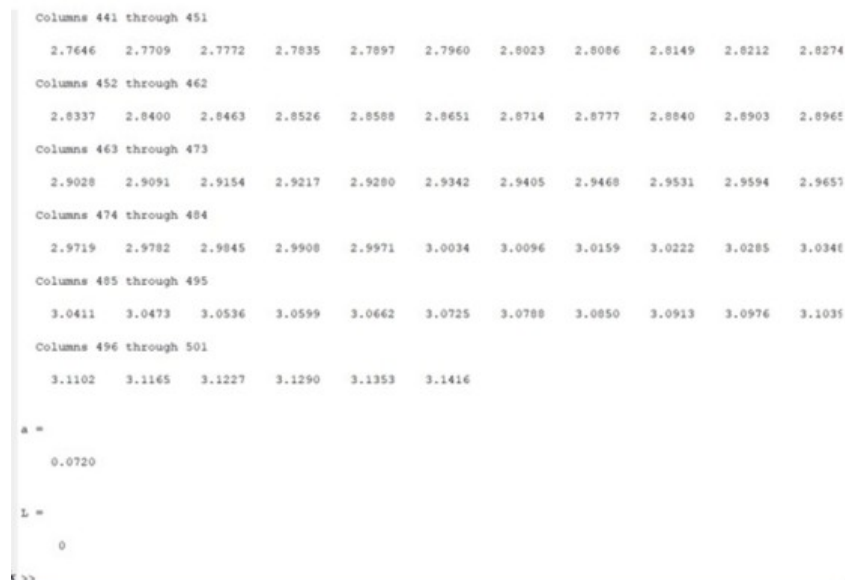


Figure 1: Amplitude Response of Type-I Linear Phase FIR Filter

MATLAB Function: Hr Type-III

Description: function computes the amplitude response of a Type-III linear phase FIR filter using sine expansion due to its odd symmetry.

```
1 function [Hr,w,c,L]= Hr_Type3(h)
2
3 M = length(h);
4 L = (M-1)/2;
5 c = [2*h(L+1:-1:1)];
6 n = 0:1:L;
7 w = (0:1:500)*pi/500;
8 Hr = sin(w*n)*c;
9 end
```

```
>> h=[1,2]

h =

     1     2

>> [Hr,w,c,L]=Hr_Type3(h)
Warning: Integer operands are required for colon operator
when used as index
> In Hr_Type3 (line 2)

Hr =

Columns 1 through 10

     0     0     0     0     0     0     0     0     0     0

Columns 11 through 20

     0     0     0     0     0     0     0     0     0     0

Columns 21 through 30

     0     0     0     0     0     0     0     0     0     0

Columns 31 through 40

     0     0     0     0     0     0     0     0     0     0
```

Figure 2: Amplitude Response of Type-III Linear Phase FIR Filter

MATLAB Function: Ideal Low-Pass FIR Filter

Description: function generates the impulse response of an ideal low-pass FIR filter using the sinc function.

```
1 function hd = ideal_lp(wc,M)
2
3 alpha = (M-1)/2;
4 n = 0:1:(M-1);
5 m = n - alpha + eps;
6 hd = sin(wc*m)./(pi*m);
7 end
```

```
>> hd=ideal_lp(wc,M)

hd =

Columns 1 through 6

    0.0318    0.0250   -0.0000   -0.0322   -0.0531   -0.0450

Columns 7 through 12

    0.0000    0.0750    0.1592    0.2251    0.2500    0.2251

Columns 13 through 18

    0.1592    0.0750    0.0000   -0.0450   -0.0531   -0.0322

Columns 19 through 21

   -0.0000    0.0250    0.0318
```

Figure 3: Impulse Response of Ideal Low-Pass FIR Filter

MATLAB Function: freqz_m

Description: function computes the magnitude, phase, and group delay response of a digital filter.

```
1 function [db,mag,pha,grd,w] = freqz_m(b,a)
2
3 [H,w] = freqz(b,a,1000,'whole');
4 H = H(1:501);
5 w = w(1:501);
6 mag = abs(H);
7 db = 20*log10((mag+eps)/max(mag));
8 pha = angle(H);
9 gr = grpdelay(b,a,w);
10 end
```

Question 1

Explanation: obtained responses verify correct implementation of linear phase FIR structures. Symmetric impulse responses result in linear phase characteristics, as confirmed by the constant group delay.

Question 2: FIR Filter Type Identification

Description: task identifies FIR filter types based on symmetry properties of the impulse response and visualizes their amplitude and zero locations.

```
1 clc; clear; close all;
2
```

3.0411
 3.0473
 3.0536
 3.0599
 3.0662
 3.0725
 3.0788
 3.0850
 3.0913
 3.0976
 3.1039
 3.1102
 3.1165
 3.1227
 3.1290
 3.1353
 3.1416

Figure 4: Magnitude, Phase, and Group Delay Response

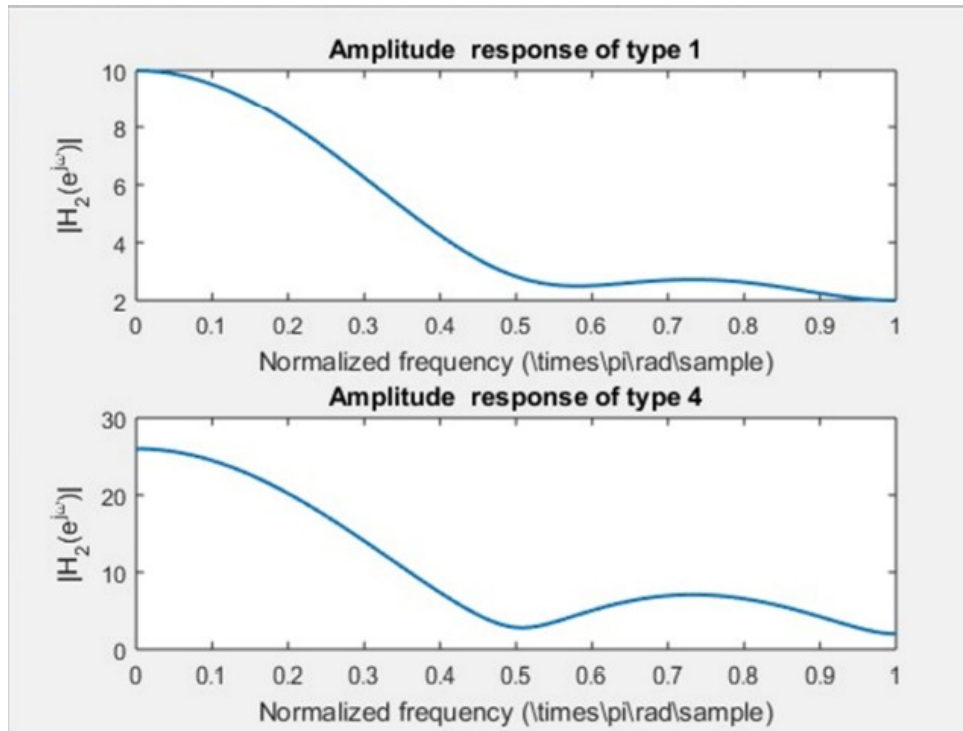


Figure 5: Verification of Linear Phase Property

3 $h_a = [-4 \ 1 \ -1 \ -2 \ 5 \ 6 \ 5 \ -2 \ -1 \ 1 \ -4];$
 4 $h_b = [-4 \ 1 \ -1 \ -2 \ 5 \ 6 \ 6 \ 5 \ -2 \ -1 \ 1 \ -4];$
 5 $h_c = [-4 \ 1 \ -1 \ -2 \ 5 \ 0 \ -5 \ 2 \ 1 \ -1 \ 4];$

```

6
7 filters = {h_a, h_b, h_c};
8 names = {'a','b','c'};
9
10 for k = 1:length(filters)
11 h = filters{k};
12 M = length(h);
13
14 '''
15 if isequal(h, fliplr(h))
16     if mod(M,2)==1
17         type = 'Type I';
18     else
19         type = 'Type II';
20     end
21 elseif isequal(h, -fliplr(h))
22     if mod(M,2)==1
23         type = 'Type III';
24     else
25         type = 'Type IV';
26     end
27 else
28     type = 'Unknown';
29 end
30
31 fprintf('Filter %s: %s Linear-Phase FIR\n', names{k}, type);
32 '''
33
34 end

```

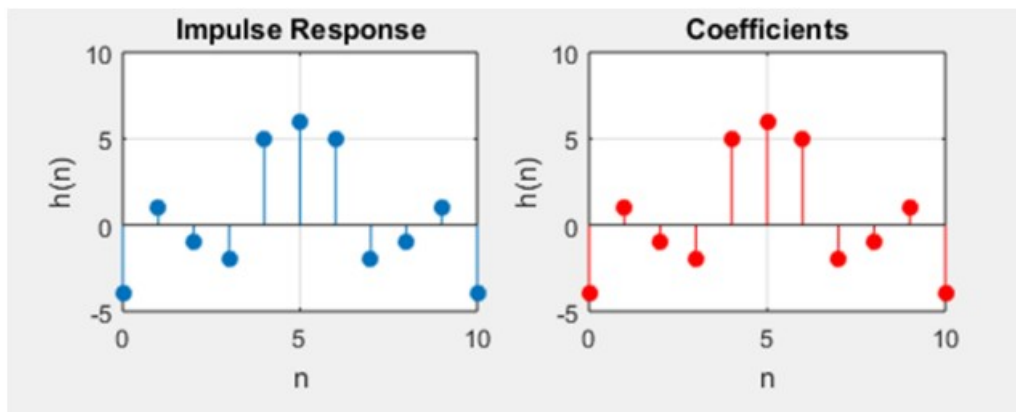


Figure 6: FIR Filter Type Identification and Analysis

Conclusion

In this lab, linear phase FIR filters were implemented and analyzed using MATLAB. Type-I and Type-III filters were studied based on symmetry properties. The frequency response, phase linearity, and zero locations validated theoretical expectations. This lab enhanced understanding of FIR filter design and classification techniques.